

THE DIM7 SYMMETRY

There are only 3 unique diminished 7th sounds in all of Western music. The other 9 dim7 chord names are inversions of these same 3 sets of notes.

THE 3 UNIQUE DIM7 CHORD GROUPS

GROUP 1

C°7

C – E $\flat$ – G $\flat$ – A

C°7	= C E $\flat$ G $\flat$ A
E$\flat$°7	= E $\flat$ G $\flat$ A C
G$\flat$°7	= G $\flat$ A C E $\flat$
A°7	= A C E $\flat$ G $\flat$

GROUP 2

C#°7

C# – E – G – B $\flat$

C#°7	= C# E G B $\flat$
E°7	= E G B $\flat$ C#
G°7	= G B $\flat$ C# E
B$\flat$°7	= B $\flat$ C# E G

GROUP 3

D°7

D – F – A $\flat$ – B

D°7	= D F A $\flat$ B
F°7	= F A $\flat$ B D
A$\flat$°7	= A $\flat$ B D F
B°7	= B D F A $\flat$

PERFECTLY SYMMETRICAL

4 equal stacked minor 3rds = 3 semitones each × 4 = 12 semitones = exact octave. The octave divides into 4 equal parts.

$$\text{m3} + \text{m3} + \text{m3} + \text{m3} = \text{P8}$$

WHAT THIS MEANS IN PRACTICE

You're memorizing 3, not 12

If you know C°7, E $\flat$ °7, G $\flat$ °7, and A°7 are all the same sound — you only need to learn the fingering once and rotate it. 12 dim7 chord names → 3 unique sounds.

Rootless dominant 7 $\flat$ 9

Every dim7 chord implies a dominant 7 $\flat$ 9. G#°7 = E7 $\flat$ 9 without the E. This is why dim7 chords resolve like dominants — they ARE dominants, just without the root stated.

4 chord tones = 4 possible dominants

C°7 can function as C7 $\flat$ 9, E $\flat$ 7 $\flat$ 9, G $\flat$ 7 $\flat$ 9, or A7 $\flat$ 9 depending on which note you treat as the leading tone. One voicing, four contextual uses.

Inversions sound identical

Because the intervals are perfectly equal, every inversion of a dim7 chord sounds like a root position chord in a different key. No inversion sounds "different" — they're all equally valid root positions.